A LEVELWISE QUASI-ISOMORPHISM OF COMPLEXES OF COMPLEXES INDUCES A QUASI-ISOMORPHISM OF PRODUCT TOTALIZATIONS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author writes '(We omit the verification that the differentials agree.)' inside the counted proof span; that sentence is kept verbatim. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The cone construction on a map of complexes and its functoriality. invoked at proof lines 30152-30155. Produces the levelwise cones whose totalization is computed.
- The product total complex of a double complex and its degreewise description. used at proof lines 30150-30151 and 30159-30165. Is the construction being compared, and the place where the degreewise product splitting occurs.
- Acyclicity of a product totalization whose rows are acyclic. invoked at proof lines 30170-30172. Is the criterion that finishes the proof once the cone identification is made.

Source reference. Original lemma 014D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1468–1468. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 30096: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30096–30096. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let*

$$(A_0^\bullet \to A_1^\bullet \to A_2^\bullet \to \ldots) \longrightarrow (B_0^\bullet \to B_1^\bullet \to B_2^\bullet \to \ldots)$$

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be a map between two complexes of complexes of abelian groups. Set $A^{p,q} = A^q_p$, $B^{p,q} = B^q_p$ to obtain double complexes. Let $Tot_\pi(A^{\bullet,\bullet})$ and $Tot_\pi(B^{\bullet,\bullet})$ be the product total complexes associated to the double complexes. If each $A^\bullet_p \to B^\bullet_p$ is a quasi-isomorphism, then $Tot_\pi(A^{\bullet,\bullet}) \to Tot_\pi(B^{\bullet,\bullet})$ is a quasi-isomorphism.

Proof. Recall that $\mathrm{Tot}_\pi(A^{\bullet,\bullet})$ in degree n is given by $\prod_{p+q=n} A^{p,q} = \prod_{p+1=n} A^q_p$. Let $C^\bullet_p$ be the cone on the map $A^\bullet_p \to B^\bullet_p$, see Derived Categories, Section 014D. By the functoriality of the cone construction we obtain a complex of complexes

$$C^\bullet_0 \to C^\bullet_1 \to C^\bullet_2 \to \ldots$$

Then we see $\mathrm{Tot}_\pi(C^{\bullet,\bullet})$ in degree n is given by

$$\prod_{p+q=n} C^{p,q} = \prod_{p+q=n} C^q_p = \prod_{p+q=n} (B^q_p \oplus A^{q+1}_p) = \prod_{p+q=n} B^q_p \oplus \prod_{p+q=n} A^{q+1}_p$$

We conclude that $\mathrm{Tot}_\pi(C^{\bullet,\bullet})$ is the cone of the map $\mathrm{Tot}_\pi(A^{\bullet,\bullet}) \to \mathrm{Tot}_\pi(B^{\bullet,\bullet})$ (We omit the verification that the differentials agree.) Thus it suffices to show $\mathrm{Tot}_\pi(A^{\bullet,\bullet})$ is acyclic if each $A^\bullet_p$ is acyclic. This follows from Lemma 30096. □